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ON THE APPLICATION OF WESENHEIT FUNCTION IN DERIVING DISTANCE TO GALACTIC CEPHEIDS

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ABSTRACT

In this paper, we explore the possibility of using the Wesenheit function to derive individual distances to Galactic Cepheids, as the dispersion of the reddening-free Wesenheit function is smaller than the optical period–luminosity (P-L) relation. When compared to the distances from various methods, the averaged differences between our results and published distances range from −0.061 to 0.009, suggesting that the Wesenheit function can be used to derive individual Cepheid distances. We have also constructed Galactic P-L relations and selected Wesenheit functions based on the derived distances. A by-product from this work is the derivation of Large Magellanic Cloud distance modulus when calibrating the Wesenheit function. It is found to be 18.531 ± 0.043 mag.

Key words: distance scale – stars: variables: Cepheids

Online-only material: color figures, machine-readable table

1. INTRODUCTION

Period–luminosity (P-L, also known as the Leavitt law) relations based on classical Cepheids in our Galaxy, referred to as the Galactic Cepheids, are important in distance scale work as well as in constraining the stellar pulsation and evolution calculations. Determining the P-L relations for Galactic Cepheids requires the measurement of distance to individual Cepheids. In contrast to extragalactic Cepheids where the distance to the host galaxy can be obtained via independent means, there are only a limited number of methods to determine distances to Galactic Cepheids (for example, see Stothers 1983; Feast & Walker 1987; Walker 1988; Wilson et al. 1991; Feast 1999, 2003; Di Benedetto 2002; Fouqué et al. 2003, 2007; Tammann et al. 2003; Ngeow & Kanbur 2004; Turner 2010, and references therein). These methods include: (1) direct parallax measurements from *Hipparcos* and *Hubble Space Telescope* (*HST*); (2) main-sequence fitting to the open clusters or associations that host Cepheids; (3) Baade–Wesselink (BW) expanding photosphere techniques that combine the integration of radial velocity and angular diameter displacements measured from surface brightness relations, interferometric measurements or semi-theoretical approach; and (4) a statistical parallax method that utilizes motions of Cepheids along the Galactic plane. Once the distances to a number of Galactic Cepheids have been measured using these methods, or a combination of them, Galactic P-L relations and the period–luminosity–color (P-L-C) relation can be calibrated.

The calibrated P-L and P-L-C relations can then be used to derive distances to a larger number of Galactic Cepheids. This in turn can be used, for example, to investigate the structure and kinematics of our Galaxy (see, for example, Stibbs 1956; Kraft & Schmidt 1963; Fernie 1968; Takase 1970; Efremov et al. 1981; Caldwell & Coulson 1987; Opolski 1988; Pont et al. 1994; Zhu 1999; Majaess et al. 2009) and mapping out the Galactic metallicity gradient or distribution (see, for example, Giridhar 1983; Andrievsky et al. 2002, 2004; Kovtyukh et al. 2005; Luck et al. 2006, 2011; Yong et al. 2006; Lemasle et al. 2007, 2008; Pedicelli et al. 2009; Luck & Lambert 2011). However, values of extinction need to be assumed or adopted for the individual Cepheids before deriving their distances using the P-L and/or P-L-C relations.

In this paper, we examine the possibility of using the Wesenheit function (Brodie & Madore 1980; Madore 1982; Moffett & Barnes 1986; Madore & Freedman 1991; Kovacs & Jurcsik 1997; Caputo et al. 2000; Kovács & Walker 2001; Leonard et al. 2003; Ngeow & Kanbur 2005; Fiorentino et al. 2007; Bono et al. 2008, 2010) to derive distances to individual Galactic Cepheids. This is because the intrinsic dispersion of P-L relations is on the order of ∼0.2 mag in the optical, hence the distance measured using the P-L relations will suffer a systematic error of the same order. In contrast, the dispersion of the Wesenheit function is greatly reduced (Fiorentino et al. 2007; Madore & Freedman 2009), as has been shown from the Large Magellanic Cloud (LMC) Cepheids: it is approximately two to three times smaller compared to the optical P-L relations (Tanvir 1999; Udalski et al. 1999; Fouqué et al. 2007; Soszyński et al. 2008a; Ngeow et al. 2009). This can reduce the systematic error in derived distances. Furthermore, in order to correct for extinction, application of both the P-L and P-L-C relations requires the estimation or determination of $E(B - V)$ values for each Cepheid. On the other hand, the Wesenheit function is reddening-free by construction (Freedman 1988; Madore & Freedman 1991). Hence the total systematic error of the derived distance does not include the extinction error when using the Wesenheit function.

Using the Wesenheit function to derive distance to Galactic Cepheids was suggested by Opolski (1983). The calibration of the Wesenheit function given in Opolski (1983), however, was based on 19 Cepheids located in open clusters or associations, by setting the distance modulus of Hyades to be 3.31 mag, with a rather large dispersion of 0.158 mag. Since then, large sets of photometric data from modern CCD measurements have become available for the Galactic Cepheids. In addition, better independent distance measurements to a larger number of Galactic Cepheids, using varied techniques as mentioned previously, are available in the literature (for example, high quality parallax measurements are available for 10 Cepheids based on *HST* observations, see Benedict et al. 2007). Therefore, the goal of this work is to examine the use of the Wesenheit function in deriving distances to Galactic Cepheids by taking advantage of these latest developments (see Fiorentino et al. 2007, for a similar approach).

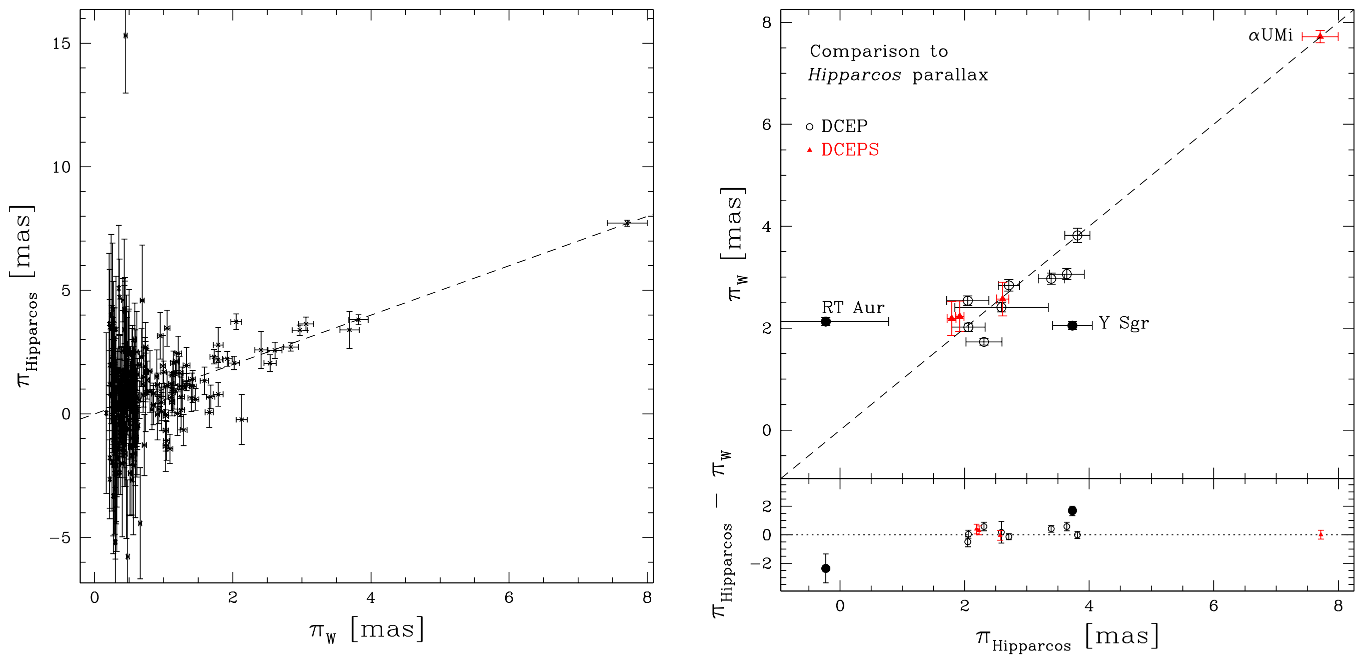

Figure 1. Comparison of the *Hipparcos* parallaxes with parallaxes converted from distance moduli calculated using Equation (1), denoted as π_W, for 223 common Cepheids in the two samples (left panel) and for the Cepheids listed in Table 2 of van Leeuwen et al. (2007). DCEP and DCEPS Cepheids are represented by open circles and filled triangles, respectively. The dashed and dotted lines in the plots are for the case of $y = x$ and $y = 0$, respectively, and not the fit to the data. Two discrepant Cepheids are marked with filled circles.

(A color version of this figure is available in the online journal.)

Table 1
Distance to Galactic Cepheids Using the Wesenheit Function[a]

Name	Type	log(P)	B	V	R	I	J	H	K	$E(B-V)$	[Fe/H]	μ_W
U AQL	DCEP	0.846	7.477	6.430	5.829	5.278	999.99	999.99	999.99	0.381	0.17	8.934
SZ AQL	DCEP	1.234	10.041	8.630	7.824	7.063	5.892	5.369	5.149	0.559	0.24	11.361
TT AQL	DCEP	1.138	8.424	7.131	6.410	5.718	4.690	4.208	4.014	0.493	0.22	9.937
...	...	...	...	...	...	...	...	...	...	...	...	...

Notes. [a] 999.99 indicates no data for a given entry.

(This table is available in its entirety in a machine-readable form in the online journal. A portion is shown here for guidance regarding its form and content.)

2. DISTANCES TO GALACTIC CEPHEIDS USING THE WESENHEIT FUNCTION

The Wesenheit function can be defined in various bandpass and filter combinations (for example, see Ngeow & Kanbur 2005, and references therein). In this paper, we only adopt the Wesenheit function in the form of $W = I - 1.55(V - I)$, because it has been demonstrated that the VI-band-based Wesenheit function is insensitive to metallicity (see, for example, Majaess et al. 2011), although an opposite result has been found in Storm et al. (2011b). Nevertheless, as discussed in Section 4, we assume that the adopted Wesenheit function is insensitive to metallicity. The Wesenheit function used in this work is adopted from Ngeow et al. (2009): $W = -3.313 \log(P) + 15.892$, with a dispersion of 0.069 mag. This Wesenheit function is derived from ∼1500 LMC fundamental mode Cepheids. Therefore, the intercept needs to be calibrated. This is equivalent to deriving the distance modulus to LMC. Assuming that our Wesenheit function is applicable to Galactic Cepheids, then the 10 Galactic Cepheids with parallax measurements from *HST* (Benedict et al. 2007) can used to calibrate the Wesenheit function. Based on these calibrators, the distance modulus of LMC was found to be 18.531 ± 0.043 mag. Therefore, distance moduli to Galactic Cepheids can be found using the following equation:

$$\mu_W = I - 1.55(V - I) + 3.313 \log(P) + 2.639, \quad (1)$$

where one only needs to know the pulsating period, P, and the VI-band intensity mean magnitudes (the I band is in the Cousin system) for a given Cepheid. The error on μ_W is taken to be $\sigma_W = \sqrt{(0.069)^2 + (0.043)^2} = 0.081$ mag.

A sample of Galactic Cepheids that have VI-band intensity mean magnitudes was compiled from Berdnikov et al. (2000), with updated intensity mean magnitudes adopted from van Leeuwen et al. (2007). Additional Cepheids or the missing VI-band intensity mean magnitudes were added from the following sources: Groenewegen (1999), Lanoix et al. (1999), Sandage et al. (1999), Tammann et al. (2003), Fouqué et al. (2007), and van Leeuwen et al. (2007). In addition, B- and R-band (in the Cousin system) intensity mean magnitudes were available for 387 and 334 Cepheids, respectively, from the cited reference. Finally, JHK-band intensity mean magnitudes in the SAAO (South African Astronomical Observatory) system are available for 229 Cepheids from van Leeuwen et al. (2007). They were converted to 2MASS (Two-Micron All Sky Survey, Skrutskie et al. 2006) systems using the color transformation

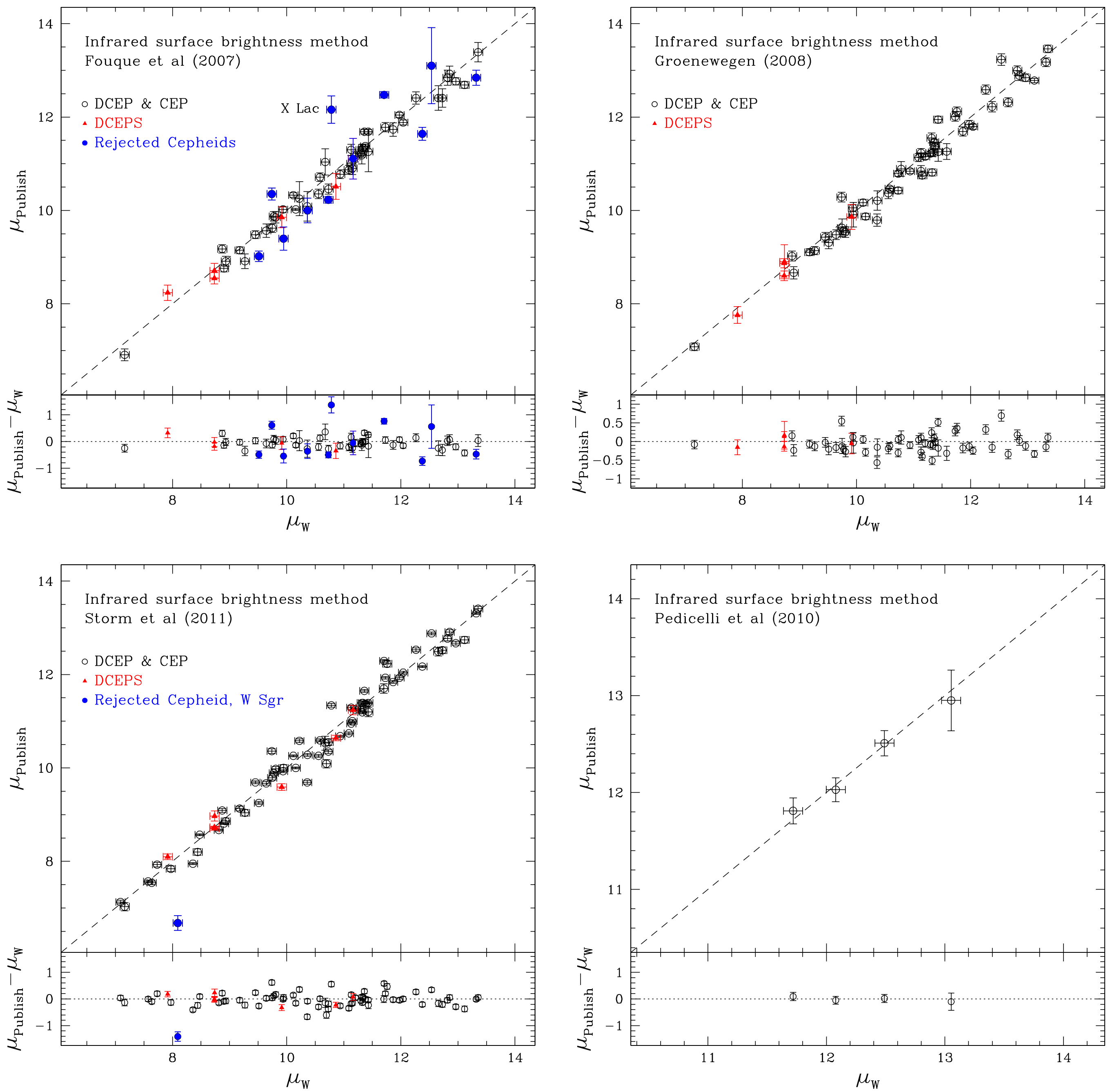

Figure 2. Comparison of the distance moduli based on the IRSB method with distance moduli measured using Equation (1) for four different IRSB samples. DCEP and CEP Cepheids are represented by open circles, while DCEPS Cepheids are represented by filled triangles. Excluded Cepheids are labeled with filled circles. The dashed and dotted lines in upper and lower panels are for the case of $y = x$ and $y = 0$, respectively, and not the fit to the data.

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equations given on the 2MASS Web site.[1] According to GCVS (General Catalog of Variable Stars, Samus et al. 2009), our sample includes 322 Cepheids that are classified as DCEP,[2] 34 of them are classified as CEP and 37 are classified as DCEPS. The *BVRIJHK*-band intensity mean magnitudes and the distance moduli calculated from Equation (1) for these Cepheids are summarized in Table 1.

Extinction and metallicity for these Cepheids are also compiled in Table 1 when available. For extinction corrections, $E(B-V)$ values taken from Fernie et al. (1995)[3] were scaled with a scale factor suggested by Tammann et al. (2003, that is, $E(B-V) = 0.951 \times E(B-V)_{\mathrm{Fernie}}$). $E(B-V)$ for 10 Cepheids, which do not have the entries from Fernie et al. (1995), were taken from either Fouqué et al. (2007) or van Leeuwen et al. (2007). The final adopted $E(B-V)$ values were listed in Table 1. For metallicity, [Fe/H] values are available from Luck & Lambert (2011, 254 Cepheids), Luck et al. (2011, 76 Cepheids), Romaniello et al. (2008, VW Cen & LS Pup), Andrievsky et al. (2003, X Sgr), Yong et al. (2006, HQ Car), and Fry & Carney (1997, QZ Nor). Metallicities from other

[1] http://www.ipac.caltech.edu/2mass/releases/allsky/doc/sec6_4b.html, which are updated transformation equations from Carpenter (2001).

[2] Both V1154 Cyg (Szabó et al. 2011) and FF Aql (Marengo et al. 2010) are updated to DCEP in this work.

[3] Available at http://www.astro.utoronto.ca/DDO/research/cepheids/

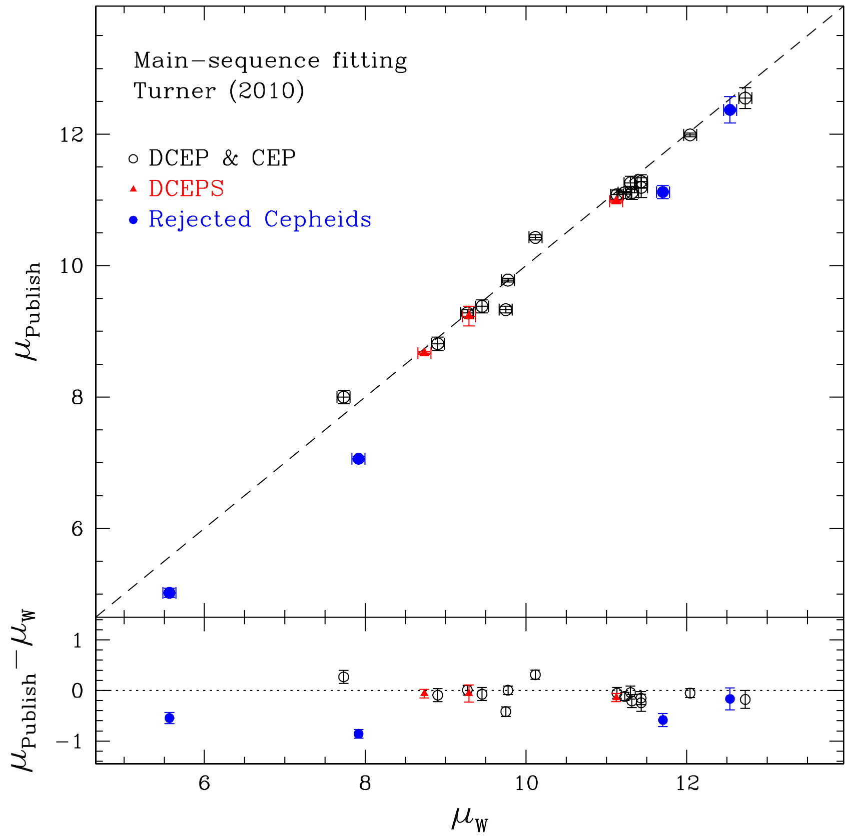

Figure 3. Comparison of the distance moduli based on the main-sequence fitting method compiled in Turner (2010) with distance moduli measured using Equation (1). DCEP and CEP Cepheids are represented by open circles, while DCEPS Cepheids are represented by filled triangles. Excluded Cepheids are labeled with filled circles. The dashed and dotted lines in upper and lower panels are for the case of $y = x$ and $y = 0$, respectively, and not the fit to the data.

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Table 2
Mean Difference of Metallicity between Other Publications and Luck & Lambert (2011)

Data Set	$N_{\rm common}$	$\langle\Delta\rangle$[a]	σ[b]
Luck et al. (2011)	180	0.07	0.08
Andrievsky et al. (2003)	48	0.08	0.08
Romaniello et al. (2008)	25	0.11	0.11
Yong et al. (2006)[c]	18	0.28	0.13
Fry & Carney (1997)[d]	10	0.19	0.09

Notes.
[a] Δ is the [Fe/H] values in Luck & Lambert (2011) minus published data sets.
[b] σ is the dispersion of the mean value.
[c] Two discrepant Cepheids, CI Per and OT Per, are excluded in the calculation.
[d] A discrepant Cepheid, EV SCT, is excluded in the calculation.

publications are transformed to the Luck & Lambert (2011) system by calculating the mean difference of the metallicity for common Cepheids, and the results are summarized in Table 2. The transformed metallicity and those from Luck & Lambert (2011) are listed in Table 1 (when available). Metallicity for Polaris is available in Andrievsky et al. (1994), though it is not included in Table 1.

2.1. Comparison to Published Results

To validate the use of Wesenheit function in deriving distance to individual Galactic Cepheids, distance moduli calculated from Equation (1) can be compared to Galactic Cepheids that possess independent distance measurements available recently in literature.

We first compared our distances to the Cepheids that have *Hipparcos* parallax measurements. Comparison of the parallaxes, measured in milliarcsecond (mas), for 223 common Cepheids listed in van Leeuwen et al. (2007) and Table 1 is shown in left panel of Figure 1. For the 14 Cepheids listed in Table 2 of van Leeuwen et al. (2007), the right panel of Figure 1 presents the comparison of *Hipparcos* parallaxes and the parallaxes based on distance moduli calculated using Equation (1). van Leeuwen et al. (2007) pointed out that Y Sgr shows a discrepancy between the *Hipparcos* parallax and the *HST* parallax given in Benedict et al. (2007). Another Cepheid that shows a large discrepancy between *Hipparcos* and *HST* parallax is RT Aur (−0.23 mas versus 2.40 mas). Our parallaxes of 2.05 ± 0.08 mas and 2.13 ± 0.08 mas for Y Sgr and RT Aur, respectively, favor the parallax measured from *HST*. Two additional Cepheids, β Dor and T Vul, show a difference of 0.58 mas between *Hipparcos* parallaxes and our parallaxes. These four Cepheids were excluded in the comparison. On the other hand, excellent agreement has been found between *Hipparcos* (7.72 ± 0.12 mas) and our parallax (7.71 ± 0.29 mas) for Polaris (α UMi). Parallaxes from *Hipparcos* were converted to distance moduli with Lutz–Kelker corrections given in Table 2 of van Leeuwen et al. (2007), and compared to the distance moduli given in Table 1. The weighted mean difference[4] of these 10 Cepheids is -0.014 ± 0.056, with a standard deviation (σ) of 0.229.

[4] Throughout the paper, difference in distance is referred to as published distance minus the distance given in Table 1.

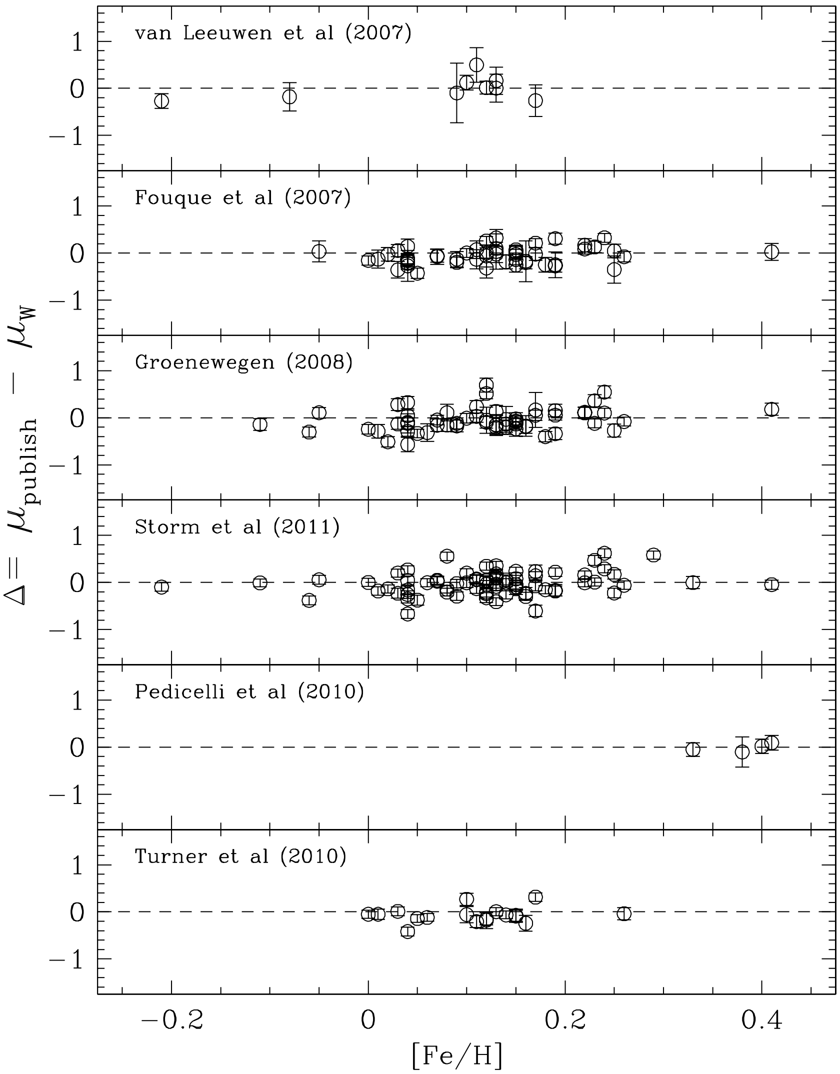

Figure 4. Difference in distance moduli for individual Cepheids as a function of [Fe/H], for the six difference samples considered in this work when comparing the published distance moduli to the distance moduli calculated from Equation (1). The dashed horizontal lines are cases for $y = 0$ and not the fit to the data.

Latest Cepheid distances using the BW infrared surface brightness (IRSB) method have been published in Fouqué et al. (2007), Groenewegen (2008), and Storm et al. (2011a). For the Fouqué et al. (2007) sample, 10 Cepheids were excluded, either due to having low quality distance measurements or because they were rejected by Fouqué et al. (2007).[5] X Lac was also excluded because the distance modulus for this Cepheid (12.159 ± 0.293) is very different from the distance modulus given in Table 1 (10.783), or from Groenewegen (2008, 10.891 ± 0.157). The weighted mean difference of the distance moduli for this sample is -0.025 ± 0.019 ($\sigma = 0.190$). For the Groenewegen (2008) sample, the weighted mean difference is -0.055 ± 0.016 ($\sigma = 0.244$). For the Storm et al. (2011a) sample, after excluding W Sgr (which is also excluded in Storm et al. 2011a) that shows a large difference between the *HST* parallax distance and the distance from IRSB, the weighted mean difference is -0.018 ± 0.011 ($\sigma = 0.242$). Finally, IRSB distances to four metal-rich Cepheids are available from Pedicelli et al. (2010), with a negligible weighted mean difference of 0.009 ± 0.085 ($\sigma = 0.088$). Comparisons of the distance moduli for these four samples are shown in Figure 2.

Recent Cepheid distance measurements based on main-sequence fitting to open clusters and associations that host Cepheids have been compiled in Turner (2010). Three Cepheids (α UMi, SU Cas, and S Vul) that show discrepancy between distances from main-sequence fitting and other distance indicators, either from *Hipparcos* parallax (for α UMi) or IRSB technique, were excluded. TW Nor was further rejected as the difference in distance modulus from Table 1 and Turner (2010) is more than 0.5 mag. For the remaining Cepheids, the weighted mean difference in distance moduli is -0.061 ± 0.025 ($\sigma = 0.167$). Figure 3 presents the comparison of the distance moduli for this sample of Cepheids.

Figure 4 shows the difference in distance moduli as a function of [Fe/H] for individual Cepheids from the samples considered previously. No obvious dependence has been found between the difference in distance moduli, calculated from Equation (1) and the published results, and the metallicity for individual Cepheids.

Using a sample of Cepheids that have [Fe/H] from Table 1, it is straightforward to derive the Galactic metallicity gradient. The Galactocentric distances were calculated using $R_G^2 = R_\odot^2 + (d\cos b)^2 - 2dR_\odot \cos b \cos l$, where $d = 10^{0.2\mu_W - 1}$ is the distance to Cepheids in kpc (with μ_W taken from

[5] These Cepheids are FM Aql, FN Aql, GT Car, BF Oph, X Pup, VZ Pup, GY Sge, YZ Sgr, SS Sct, and S Vul.

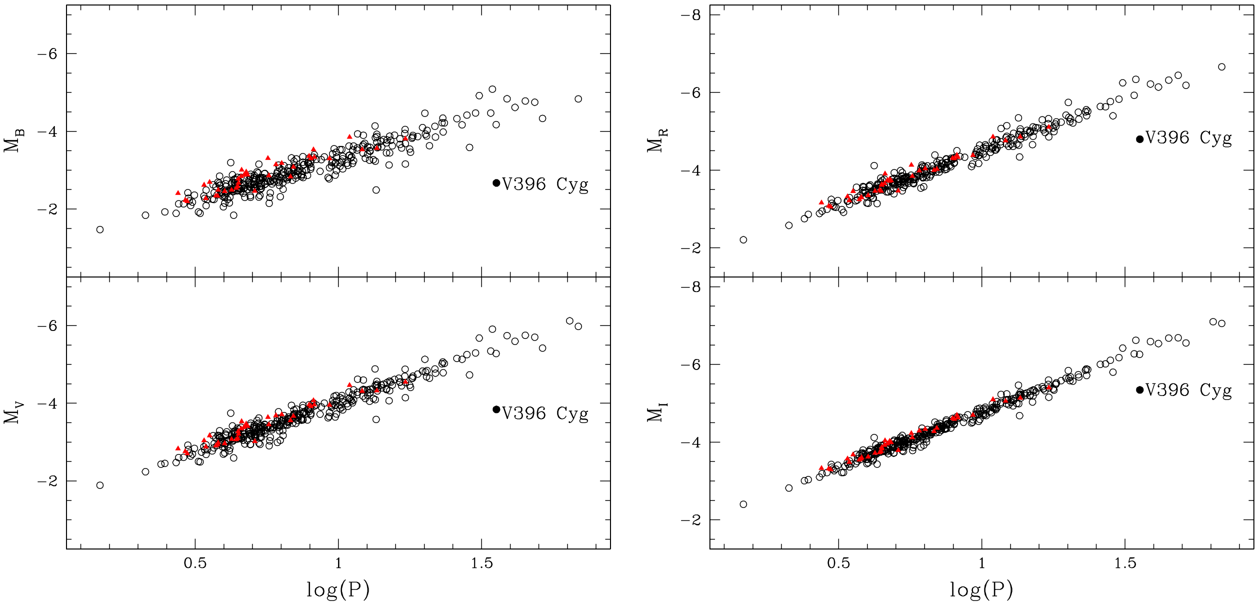

Figure 5. Extinction corrected *BVRI*-band P-L relations, based on the mean intensities and distance moduli listed in Table 1. Open circles are for Cepheids classified as DCEP and CEP, and filled (red) triangles are for DCEPS Cepheids. The outlier is marked as filled circles.

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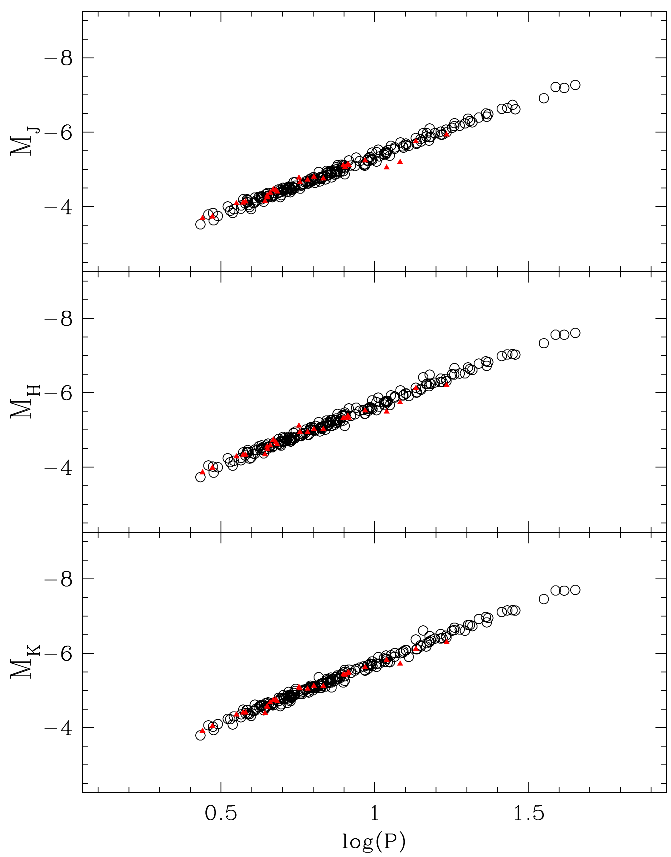

Figure 6. Same as Figure 5, but for JHK bands in the 2MASS system.

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Table 1), l and b are Galactic coordinates in radians, and $R_\odot = 7.9$ kpc is the Galactocentric of the Sun (McNamara et al. 2000). The resulting metallicity gradient is [Fe/H] $= 0.580(\pm0.024) - 0.058(\pm0.003)R_G$, with a dispersion of 0.101, which is consistent with the relation found by Luck & Lambert (2011).

3. THE GALACTIC PERIOD–LUMINOSITY RELATIONS AND WESENHEIT FUNCTIONS

It is straightforward to derive the Galactic P-L relations using the data and distance moduli given in Table 1. For the 28 Cepheids that do not have them, $E(B-V)$ values are not used in deriving the P-L relations. Total-to-selective absorption ratios were adopted from Fouqué et al. (2007) as follows: $R_{\{B,\ V,\ R,\ I,\ J,\ H,\ K\}} = \{4.23, 3.23, 2.73, 1.96, 0.94, 0.58, 0.38\}$. Finally, *Spitzer* IRAC- and MIPS-band photometry were available from Marengo et al. (2010) for 29 Cepheids. Extinction is ignored for the mid-infrared-band photometry, as it is negligible at these wavelengths (Freedman et al. 2008, 2011; Ngeow & Kanbur 2008; Ngeow et al. 2009; Freedman & Madore 2010). The 70 μm-band photometric data are not considered in this work, as most of them only have upper limits in flux. The multi-band P-L relations are presented in Figures 5–8.

A clear outlier, V396 Cyg, is shown in Figure 5. Based on its location in the P-L plane, it is possible that this Cepheid is a type II Cepheid, and not a classical population I Cepheid. The Wesenheit function can be used to disentangle type II Cepheids from the classical Cepheids (for example, see Soszyński et al. 2008b, their Figure 1). For Wesenheit functions in the form of $W_{BV} = V - 3.23(B-V)$ and $W_{RI} = I - 2.55(R-I)$, the Wesenheit magnitudes of this outlier are $\sim$0.69 mag($= 4.1\sigma$, where σ is the dispersion of the fitted Wesenheit function) and $\sim$0.36 mag($= 2.7\sigma$), respectively, fainter from the ridge line of the fitted Wesenheit function given in Table 4. Therefore, it is inconclusive whether this outlier should be a type II or classical Cepheid. Another possible explanation is that the published extinction value for this outlier, $E(B-V) = 1.092$, is underestimated. A higher value of $E(B-V) \sim 1.5$ can provide better agreement for its absolute magnitudes to other Cepheids with similar periods. Detailed investigation of this Cepheid is beyond the scope of this paper, but nevertheless it is clear that this Cepheid should be excluded when fitting the P-L relations. For remaining Cepheids, the fitted multi-band P-L

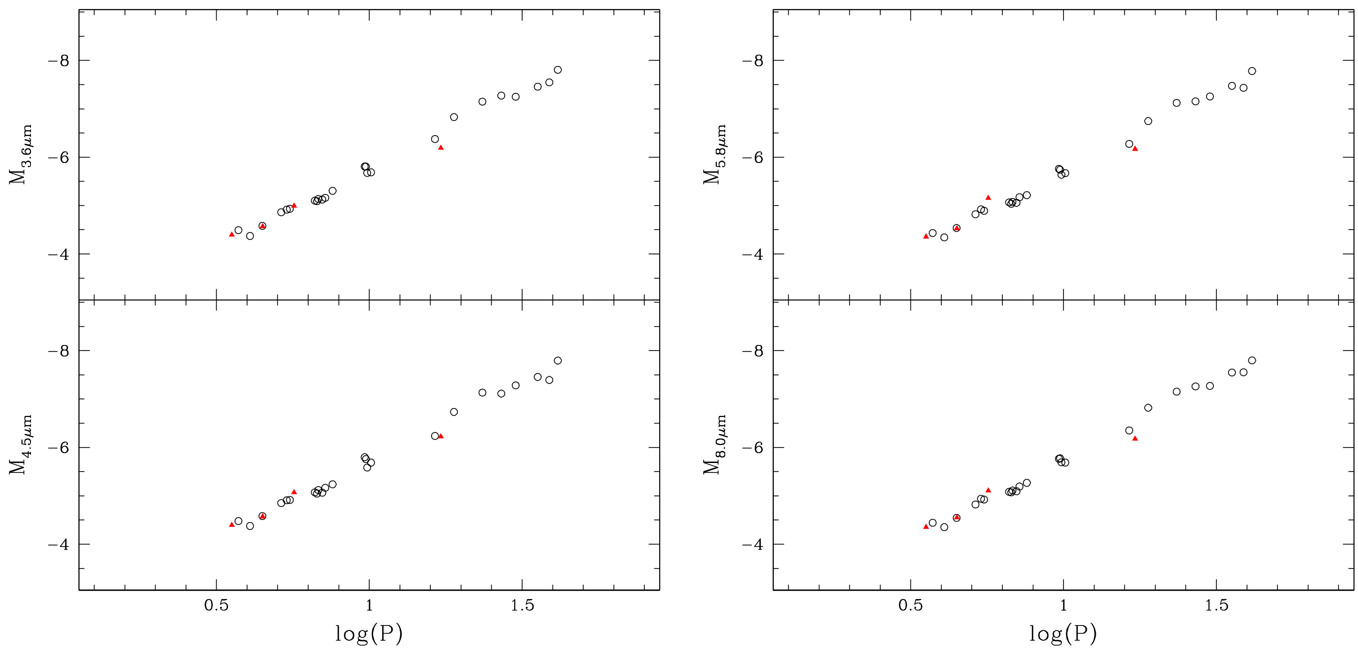

Figure 7. Same as Figure 5, but for *Spitzer* IRAC bands.
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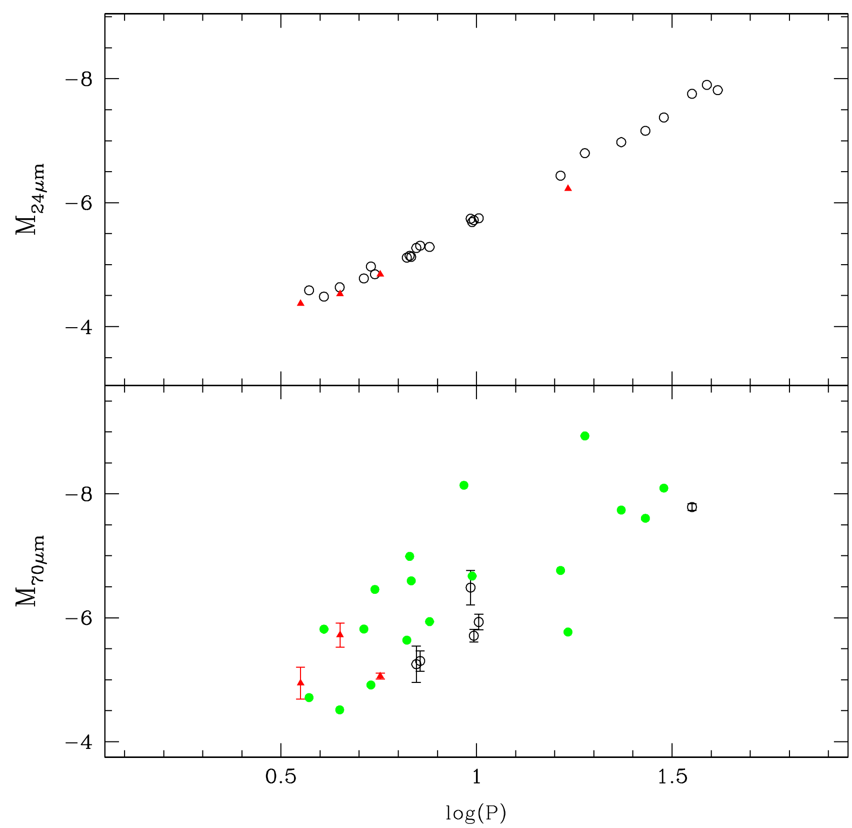

Figure 8. Same as Figure 5, but for *Spitzer* MIPS bands. Cepheids that only have the upper flux limits (lower limits in magnitudes) in MIPS 70 μm are represented as filled (green) circles.
(A color version of this figure is available in the online journal.)

relations are summarized in Table 3. Slopes and intercepts of these P-L relations as a function of wavelengths were presented in Figure 9. Both the P-L slopes and intercepts monotonically decrease from B to K band (except for the J-band P-L slope with DCEPS Cepheids), and approach an asymptotic value in the mid-infrared. The "dip" around 4.5 μm- and 5.8 μm-band P-L slopes is interpreted as being due to the CO absorption that affects this band (Marengo et al. 2010).

Data in Table 1 can also be used to derive the Galactic Wesenheit functions in other bandpass and color combinations

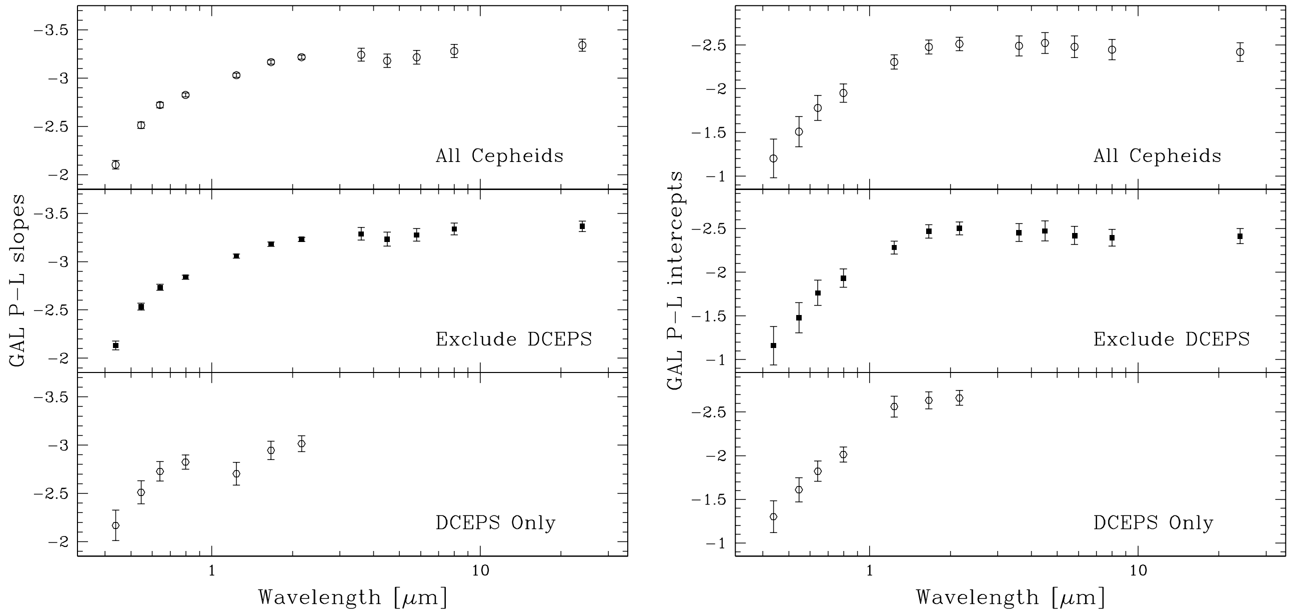

Figure 9. P-L slopes (left panels) and intercepts (right panels) as a function of wavelength based on the results presented in Table 3.

Table 3
The Galactic P-L Relations[a]

Band	N	a_λ	b_λ	σ
		All Cepheids		
B	357	-2.103 ± 0.044	-1.201 ± 0.039	0.221
V	364	-2.513 ± 0.033	-1.508 ± 0.030	0.173
R	319	-2.722 ± 0.029	-1.781 ± 0.026	0.143
I	364	-2.826 ± 0.020	-1.951 ± 0.018	0.105
J	229	-3.030 ± 0.022	-2.306 ± 0.020	0.082
H	229	-3.166 ± 0.022	-2.478 ± 0.020	0.080
K	229	-3.217 ± 0.021	-2.513 ± 0.019	0.077
3.6 μm	29	-3.242 ± 0.067	-2.491 ± 0.070	0.115
4.5 μm	29	-3.180 ± 0.070	-2.523 ± 0.073	0.120
5.8 μm	29	-3.216 ± 0.072	-2.480 ± 0.074	0.123
8.0 μm	29	-3.280 ± 0.068	-2.447 ± 0.071	0.117
24 μm	29	-3.341 ± 0.062	-2.420 ± 0.064	0.107
		Exclude DCEPS		
B	320	-2.132 ± 0.045	-1.159 ± 0.041	0.219
V	327	-2.534 ± 0.035	-1.478 ± 0.032	0.173
R	282	-2.734 ± 0.031	-1.764 ± 0.029	0.145
I	327	-2.839 ± 0.021	-1.932 ± 0.019	0.105
J	203	-3.058 ± 0.021	-2.282 ± 0.019	0.073
H	203	-3.181 ± 0.022	-2.467 ± 0.020	0.077
K	203	-3.231 ± 0.021	-2.501 ± 0.020	0.075
3.6 μm	24	-3.289 ± 0.065	-2.454 ± 0.069	0.103
4.5 μm	24	-3.233 ± 0.072	-2.472 ± 0.077	0.114
5.8 μm	24	-3.277 ± 0.065	-2.420 ± 0.070	0.104
8.0 μm	24	-3.337 ± 0.061	-2.395 ± 0.065	0.096
24 μm	24	-3.366 ± 0.054	-2.413 ± 0.058	0.086
		DCEPS Only		
B	37	-2.168 ± 0.158	-1.302 ± 0.120	0.182
V	37	-2.511 ± 0.120	-1.610 ± 0.091	0.139
R	37	-2.728 ± 0.101	-1.822 ± 0.076	0.116
I	37	-2.824 ± 0.073	-2.013 ± 0.056	0.085
J	26	-2.704 ± 0.118	-2.562 ± 0.095	0.120
H	26	-2.945 ± 0.095	-2.634 ± 0.077	0.097
K	26	-3.015 ± 0.083	-2.661 ± 0.067	0.085

Notes. [a] The P-L relation takes the form of $M_\lambda = a_\lambda \log(P) + b_\lambda$, σ is the dispersion of the fitted P-L relations.

Table 4
Selected Galactic Wesenheit Functions[a]

W	a	b	σ	N
		All Cepheids		
$V - 3.23(B - V)$	-3.811 ± 0.033	-2.514 ± 0.030	0.179	384
$I - 2.55(R - I)$	-3.091 ± 0.025	-2.394 ± 0.023	0.130	333
$J - 1.68(J - K)$	-3.343 ± 0.026	-2.654 ± 0.024	0.097	229
$J - 0.74(V - I)$	-3.276 ± 0.019	-2.615 ± 0.017	0.071	229
$H - 0.46(V - I)$	-3.320 ± 0.021	-2.672 ± 0.019	0.077	229
$K - 0.30(V - I)$	-3.317 ± 0.020	-2.639 ± 0.018	0.074	229
$H - 0.41(V - I)$	-3.293 ± 0.020	-2.633 ± 0.019	0.076	229
		Exclude DCEPS		
$V - 3.23(B - V)$	-3.811 ± 0.034	-2.520 ± 0.032	0.178	347
$I - 2.55(R - I)$	-3.113 ± 0.026	-2.362 ± 0.024	0.128	296
$J - 1.68(J - K)$	-3.350 ± 0.027	-2.650 ± 0.025	0.097	203
$J - 0.74(V - I)$	-3.297 ± 0.016	-2.600 ± 0.015	0.057	203
$H - 0.46(V - I)$	-3.330 ± 0.020	-2.667 ± 0.019	0.071	203
$K - 0.30(V - I)$	-3.328 ± 0.020	-2.631 ± 0.019	0.071	203
$H - 0.41(V - I)$	-3.307 ± 0.020	-2.624 ± 0.018	0.071	203
		DCEPS Only		
$V - 3.23(B - V)$	-3.620 ± 0.154	-2.605 ± 0.117	0.177	37
$I - 2.55(R - I)$	-3.072 ± 0.094	-2.500 ± 0.071	0.108	37
$J - 1.68(J - K)$	-3.227 ± 0.100	-2.729 ± 0.081	0.102	26
$J - 0.74(V - I)$	-2.955 ± 0.120	-2.839 ± 0.097	0.122	26
$H - 0.46(V - I)$	-3.103 ± 0.094	-2.806 ± 0.094	0.096	26
$K - 0.30(V - I)$	-3.117 ± 0.078	-2.773 ± 0.063	0.080	26
$H - 0.41(V - I)$	-3.059 ± 0.096	-2.790 ± 0.077	0.098	26

Notes. [a] The Wesenheit function takes the form of $W = a \log(P) + b$, σ is the dispersion of the fitted Wesenheit functions.

(except for the VI bands). A number of selected Wesenheit functions involving $BRJHK$ bands are presented in Table 4.[6] Wesenheit function in the from of $W = H - 0.41(V - I)$, adopted by the SH0ES team (Riess et al. 2011), is also included in Table 4. For Wesenheit function involved B band, the

[6] Other forms of Wesenheit functions are straightforward to derive from Table 1, hence they are not included here.

dispersion of the relation is largest with the steepest slope, which is consistent with the results found in Bono et al. (2010). The dispersions of the JHK-band-based Wesenheit functions, on the other hand, are in the order of $\sim$0.1 and smaller, suggesting they could also be used in future distance scale work. It is worth pointing out that the Wesenheit function in the form of $W = J - 0.74(V - I)$, based on DCEP and CEP Cepheids, has a dispersion of 0.057, about 17% smaller than the VI-band-based Wesenheit function adopted in this work.

4. DISCUSSION AND CONCLUSION

In conclusion, mean differences between the distance moduli given in the literature and those calculated from Equation (1) range from negligible to about 6 percent, depending on the samples and the methods used to derive independent distances to the Galactic Cepheids. However, various assumptions have been made when deriving these independent distances. For example, certain expressions of the p-factors (to covert radial velocities to pulsational velocities) need to be adopted when applying the BW-type analysis. In contrast, Equation (1) is assumed to be applicable to Galactic Cepheids, though the relation is derived from LMC Cepheids. Good agreement between the 10 Cepheids with *Hipparcos* parallaxes and the distance calculated from Equation (1) suggested that the Wesenheit function used in this work is not sensitive, or mildly sensitive, to metallicity. This is also supported by recent empirical work presented in Bono et al. (2010, and references therein), showing that the empirical slopes of the Wesenheit functions for both metal-poor and metal-rich galaxies are consistent with the LMC Wesenheit slopes. Majaess et al. (2011) have also demonstrated that the distance moduli to Magellanic Clouds would significantly disagree with the canonical values if a strong metallicity correction to the VI-band-based Wesenheit function were adopted. In terms of theoretical works, Fiorentino et al. (2007) and Bono et al. (2008) also found a weak or mild dependence of the VI-band-based Wesenheit function on metallicity. Other advantages of using the VI-band-based Wesenheit function, in addition to being reddening-free by definition, include the smaller dispersion of the relation, being linear (Marconi et al. 2005; Ngeow & Kanbur 2005; Fiorentino et al. 2007; Madore & Freedman 2009; Ngeow et al. 2009; Bono et al. 2010), and only needing the VI-band mean magnitudes and periods of the Cepheids. Certainly, the verification of the distance moduli given in Table 1 and the use of the Wesenheit function in deriving distances relies on the parallaxes measured from *Gaia*.

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